EAST WEST UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE303: STATISTICS FOR DATA SCIENCE
SUMMER 2026 SEMESTER

PROJECT 2: VERIFIED-PURCHASE PREDICTION AND REVIEW TRUSTWORTHINESS

Research Angle: Evaluating Whether Linguistic and Behavioral Signals Distinguish Verified from Non-Verified Bangla Book Reviews on Rokomari.com

Date of Submission: 15th August 2026

Group Details			
Group Number	3	**Section**	6
Group Members		**Id**	
Mahadi Prodhan		2023-2-60-033	
Rishad Imtinan		2023-3-60-014	
Md Mahiruddin Chowdhury		2023-3-60-504	
Instructor Details			
Course Instructor		Dr. Nafis Sadeq	

1. Introduction and Motivation

Modern E-commerce bookshops like Rokomari.com rely heavily on customer reviews and ratings, either in numerical or star-based ratings to decide which books to buy. However, these online reviews can often be misleading because it being biased, promotional or written by people who never purchased the book through their online store. Furthermore, the platforms also display unverified readers' opinions and normal feedback. Ecommerce platforms are asked to keep their review or product feedback system completely trustworthy through verified customers. Finding the difference between verified and non-verified review submissions is therefore crucial for maintaining the site's integrity, search engine optimization and users trust and recommendations.

An essential ethical principle that frames this research is that non-verified reviews should not be considered fake reviews by default. Non-verified review does not necessarily mean deception. The analysis of qualitative data shows that most unverified reviewers of Rokomari are real book readers who have bought or borrowed physical copies of books from brick-and-mortar stores and libraries before writing their comments online. In this case, the better framing approach for this research is the purchase verification prediction or review trustworthiness analysis.

The main goal of the project is to investigate how behavior and language differences can help determine if a review is verified or non-verired. Every step of analysis including data filtering through exploratory analysis, hypothesis testing, multiple modeling and training, and thresholding helps us identify whether certain aspects of it can be used to build an effective automated review verification system.

2. Research Question & Problem Statement

Our central research angle of this project evaluates and determines whether linguistic and behavioral signals can differentiate between verified from non-verified Bangla book reviews on the ecommerce platform, Rokomari.com.

To achieve the objective, the problem is defined as a binary classification problem. In particular, for a given review and its accompanying metadata (e.g., review text length or word count, number of stars given, punctuation marks, upvote and downvote count for helpfulness, rating deviation and book category), the aim is to predict the binary target variable Y in {0, 1} that equals 1 if the purchase is verified, and 0 otherwise.

Identifying the problem as a binary classification problem creates a clear probabilistic decision boundary $P(y = 1 \mid x)$ using logistic regression. In this study, our one major milestone of analysis is to evaluated whether linguistic features such as word counts, punctiuation marks and behavioral features such as rating deviation and genre categories have significant predictive power to accurately picture the distinction between verified and unverified customer reviews.

3. Dataset & Target-Label Construction

Our project uses the official RokomariBG e-commerce dataset, which containsa total of 463406 rows where 149,515 are book records and over 228,000 are unique user reviews. About 55% of the reviews in the dataset include a verified-purchase indicator, which provides a solid basis for constructing ground-truth labels.

Before choosing the final target label, three candidate variables in the dataset were weighed

against each other: multi-class star ratings which range from 1 to 5 stars, helpfulness vote ratios which is positive votes divided by total votes, and lastly binary verified-purchase flags. Table 1 lays out the trade-offs between them.

Table 1: Candidate Target Variable Evaluation & Selection Decisions.

Candidate Target	Data Structure & Limitations	Selection Decision & Rationale
Star Ratings (1 to 5 Stars)	Over 80% of reviews were 5 stars	Rejected: Extreme positive skew masks model learning.
Helpfulness Vote Ratio	Over 90% of reviews had zero votes	Rejected: High voting sparsity invalidates continuous regression.
Verified Purchase Status	Clean 50/50 split (verified vs unverified)	SELECTED: Balanced, objective ground-truth target label.

As shown in Table 1, star ratings skewed heavily towards 5 stars which is over 80% while helpfulness votes mostly missing or sparse data where over 90% of them had zero votes. Binary verified purchase status provided the cleanest, most objective target label. To ensure statistical integrity, the dataset was checked for duplicate user book submissions, and exactly 538,386 duplicate records were removed to satisfy the statistical independence assumption.

Additionally, we determined how sample size affects statistical stability, two sampling trials were executed where a small trial of 1,000 reviews with a standard error of SE = 4..4247 and a full stratified trial of 20,000 reviews with 10,000 verified and 10,000 unverified. In the 20,000 sample trial, standard error dropped by over 77.5% to SE = 0.9936. This empirical result highlighted the Central Limit Theorem, demonstrating that a larger 20,000 stratified sample size provides a stable baseline for classification modeling.

4. Exploratory Data Analysis (EDA)

To find out whether the behavioral and linguistic signals in the data separate verified from non-verified reviews, a comprehensive exploratory data analysis was done.

Figure 1 shwos the target class distribution contains exactly 25,449 verified reviews (y = 1) and 18,260 non-verified reviews (y = 0), forming a 58.22% to 41.78% class split across 43,709 total observations. The star rating breakdown shows that non-verified reviewers leave 5-star ratings at a higher proportion which is 69.71% compared to verified buyers pf 62.55%, providing an initial behavioral signal.

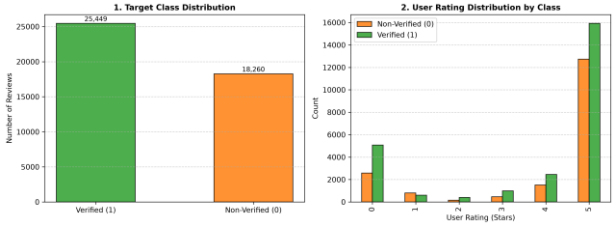


Figure 1: Target Class Distribution & Star Ratings Breakdown.

Figure 2 shows the histogram binning sensitivity experiment, comparing four layouts, which are 15 bins, 30 bins, 60 bins, and log-scale frequency. The 15-bin graph was too simple and hid distribution details, while the 60-bin layout introduced excessive noise. The 30-bin layout provided the clearest view of word count distribution.

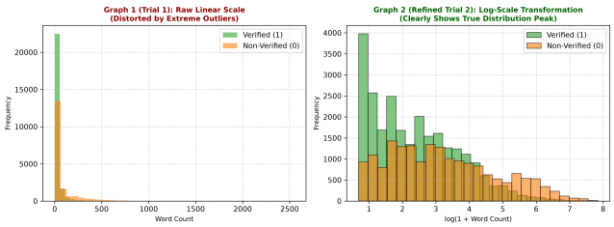


Figure 2: Histogram Binning Sensitivity Trial.

Figure 3 compares raw word counts against a log-transformed scale log(1 + x). In the raw graph, a few extremely long reviews of over 2,000 words are extremely right skewed as majority of observations are into the first bar. Applying logarithmic scaling, log(1 + x) compressed long reviews and converted the skewed distribution into a symmetric bell shape suitable for classification modeling.

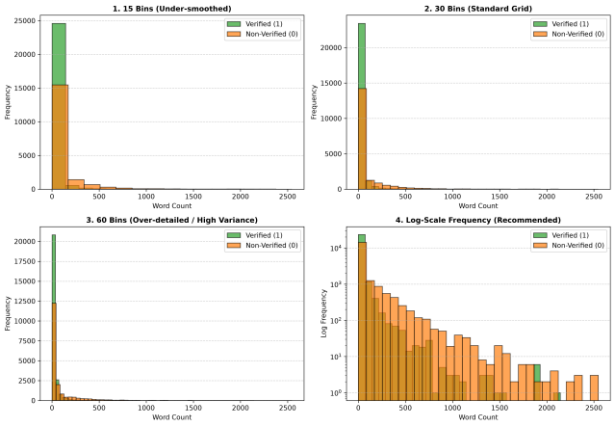


Figure 3: Raw Linear Scale vs. Log-Scale Transformation.

As shown in Figure 4, side-by-side boxplots display review word counts before and after removing upper-tail outliers beyond Q3 + 1.5 x IQR (100 words). The quantitative comparison in Table 2 highlights the difference between mean and median location metrics.

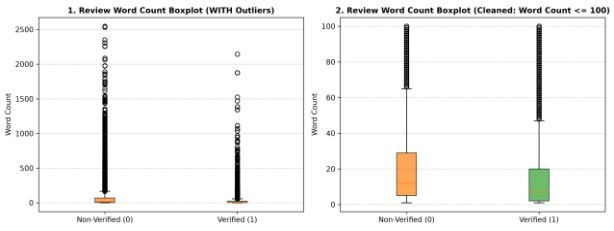


Figure 4: Review Word Count Boxplots WITH vs. WITHOUT Outliers.

Table 2: Statistical Metrics Comparison With vs. Without Outliers.

Statistical Metric	With Outliers (Raw)	Without Outliers (Cleaned <= 100)
Non-Verified Mean Word Count	89.78 words	21.12 words (-68.66 shift)
Verified Mean Word Count	28.02 words	15.56 words (-12.46 shift)
Non-Verified Median Word Count	19.00 words	12.00 words
Verified Median Word Count	8.00 words	7.00 words
Non-Verified Variance	37,020.46	525.78 (-98.6% drop)
Verified Variance	5,755.09	400.07 (-93.0% drop)
Non-Verified IQR	64.00 words (Q1: 6, Q3: 70)	24.00 words (Q1: 5, Q3: 29)
Verified IQR	22.00 words (Q1: 2, Q3: 24)	18.00 words (Q1: 2, Q3: 20)

Table 2 shows that when outliers were removed, Non-Verified Mean word count shifted from 89.78 down to 21.12 words (-68.66 shift) and sample variance dropped by 98.6%. In contrast, sample medians remained stable (Non-Verified Median moved from 19.0 to 12.0, Verified Median moved from 8.0 to 7.0). The result highlights that sample mean is sensitive to extreme outliers, whereas sample median acts as a robust metric for analysis.

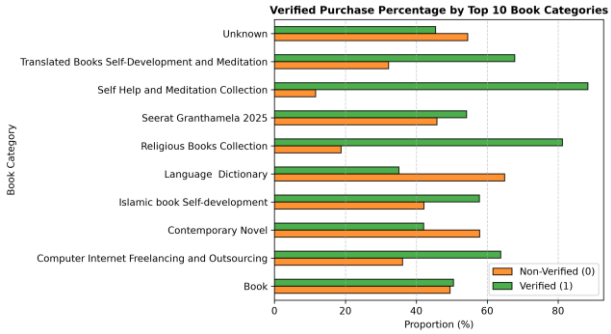


Figure 5: Top 10 Book Categories on Rokomari.com.

Figure 5 displays the top 10 book categories on Rokomari, showing that general fiction, religious collections, and self-help categories contain the largest volume of user reviews.

5. Methodology (Baseline & Proposed Models)

To test whether behavioral and linguistic signals distinguish verified reviews and address the research objective, exactly 5 distinct machine learning models were tested:

Model 1 Baseline Constant Mean Predictor: Predicts a constant sample mean probability (y_bar = 0.5822) for all observations, establishing a lower-bound baseline validation accuracy of 58.22% and BCE loss of 0.679567.

Model 2 Baseline Constant Median Predictor: Predicts a constant sample median class (median = 1) for all observations, and hits a benchmark baseline validation accuracy of 58.22%.

Model 3 Custom Vectorized L2 (Ridge) Regularized Logistic Regression: Custom NumPy gradient descent model incorporating Binary Cross-Entropy (BCE) loss and quadratic weight decay regularization of (0.5 * lambda * sum(theta_j^2)), decaying weights smoothly while keeping active features where the outputs were found to have Initial Loss = 0.693150 and Final Training Loss = 0.662313.

Model 4 Custom Vectorized L1 (Lasso) Regularized Logistic Regression: Custom NumPy gradient descent model incorporating absolute sub-gradient regularization strength (lambda * sum(|theta_j|)), performing automated feature selection by zeroing out 3 uninformative category weights where . theta_j = 0.0, Initial Loss = 0.693508 and Final Training Loss = 0.677043.

Model 5 Benchmark Scikit-Learn Logistic Regression (L-BFGS): External benchmark model using Scikit-Learn's second-order Hessian L-BFGS solver, achieving 66.34% validation accuracy to verify that the custom NumPy implementation with 59.85% validation accuracy reached comparable performance.

Before model training, three parametric statistical tests confirmed that significant behavioral differences exist between verified and unverified reviews. A two-sample Z-test for 5-star proportions gave us Z = -15.5289, p = 0.0000. A Welch's two-sample independent t-test for review length showed T = -41.1466, p = 0.0000. A one-way ANOVA across book categories produced F = 115.9362, p = 4.5458e-97, with SS_between = 43.2539 and SS_within = 1055.4539.

A single one-way ANOVA was used instead of multiple pairwise t-tests to prevent family-wise Type I error inflation. Comparing five categories with individual pairwise t-tests would require 10 separate comparisons, which pushes the cumulative Type I error probability above 40%. One-way ANOVA evaluates all category means at once and keeps the overall Type I error at 5%.

The feature pipeline built 6 continuous behavioral features (word count, character count, punctuation intensity, helpfulness ratio, rating deviation, price) and 13 binary category indicator columns, dropping 1 reference category ('Collections Box Sets'). Min-max

normalization then scaled the continuous features into [0, 1].

6. Experimental Setup

The experimental setup used a 3-way stratified data partition: 70% training set (X_train = 30,596, y_train = 30,596), 15% validation set (X_val = 6,556, y_val = 6,556), and 15% test set (X_test = 6,557, y_test = 6,557). A constant bias column x0 = 1 was added to the front of the feature matrix, which brought the total parameter dimension to 21 (θ_0 to θ_{20}).

To select optimal hyperparameters for Model 3 and Model 4, a grid search ran across 12 candidate combinations of learning rate (Alpha in {0.01, 0.1, 0.5}) and regularization strength (Lambda in {0.0, 0.01, 0.1, 1.0}), evaluated on the 15% validation set. Table 3 details the grid search results below.

Table 3: Hyperparameter Grid Search Results on Validation Set.

Learning Rate (alpha)	RegStrength (lambda)	Validation Loss	Validation Accuracy
0.01	0.00	0.679076	58.22%
0.01	0.01	0.679144	58.22%
0.01	0.10	0.679677	58.22%
0.01	1.00	0.681564	58.22%
0.10	**0.00**	**0.658178**	**59.69% (OPTIMAL)**
0.10	0.01	0.660821	59.50%
0.10	0.10	0.671587	58.22%
0.10	1.00	0.677521	58.22%
0.50	0.00	0.744899	52.41%
0.50	0.01	0.790122	47.67%
0.50	0.10	0.844596	46.48%
0.50	1.00	1.045944	46.48%

As shown in Table 3, the grid search selected Alpha* = 0.10 and Lambda* = 0.00 as the optimal hyperparameter settings, achieving the lowest validation loss of 0.658178 and 59.69% validation accuracy. Setting Lambda = 1.00 induced severe underfitting (loss spiked to 1.0459), while Alpha = 0.50 caused gradient overshooting.

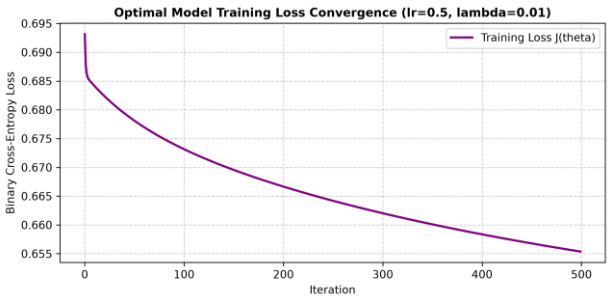


Figure 6: Optimal Model Training Loss Convergence Curve.

Figure 6 depicts the optimal model training loss convergence curve across 500 iterations, illustrating smooth monotonic decay from Initial Loss = 0.693147 down to Final Loss = 0.655366.

7. Results & Threshold Calibration

The trained proposed model was tested on the 15% unbiased test set (6,557 samples), under two decision rules: a standard threshold (T = 0.50) and a calibrated decision threshold (T* = 0.5899, derived from the validation median probability). Figure 7 and Table 4 show that the threshold calibration made a real difference to the project's results.

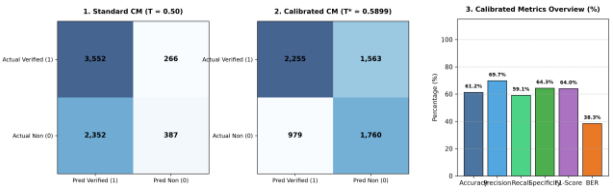


Figure 7: Test Confusion Matrices (Default 0.50 vs. Calibrated 0.5899) and Performance Bar Chart.

Table 4: Test Performance Comparison Across Evaluation Metrics (Default vs. Calibrated Threshold).

Evaluation Metric	Default Threshold (T=0.50)	Calibrated Threshold (T*=0.5899)
True Positives (TP)	3,552	**2,255**
True Negatives (TN)	387	**1,760 (+1,373 gain!)**
False Positives (FP)	2,352	**979 (-1,373 drop)**
False Negatives (FN)	266	**1,563**
Accuracy	60.07%	**61.23% (+1.16%)**
Precision	60.16%	**69.73% (+9.57% boost)**
Specificity (TNR)	14.13%	**64.26% (+50.13% boost)**
False Positive Rate (FPR)	85.87%	**35.74% (-50.13% drop)**
Balanced Error Rate (BER)	46.42%	**38.34% (-8.08% improvement)**

To evaluate whether raw accuracy is a reliable metric, we first assessed the dataset's class balance and threshold calibration. Under the default threshold (T = 0.50), raw accuracy reached 60.07%. However, this figure concealed a major classification failure on non-verified reviews: specificity was only 14.13%, and the uncalibrated Balanced Error Rate (BER) stood at 46.42%.

We then calibrated the decision threshold to T* = 0.5899, and the results showed a clear improvement. Specificity increased to 64.26% (a 50.13% boost), precision increased to 69.73% (a 9.57% boost), and BER dropped to 38.34% (an 8.08% improvement). Overall, test accuracy reached 61.23%, an 11.23% lift over random baseline guessing. At last, we determined that threshold calibration was crucial to meeting the project's primary goal.

8. Limitations

According to our results we can assume that there are three possible reasons why our model struggled to achieve correct results.

1. Overlapping Feature Distributions: Verified and non-verified reviewers often write reviews of similar length and leave similarly high ratings. This overlap in the data leads to high classification variance and limits how well a linear model can separate the two groups. 2. Linear Decision Boundary Assumption: Logistic regression assumes a linear decision boundary (theta_T * x = 0), so it cannot capture the complex, non-linear relationships between review length and rating deviation. 3. High Data Sparsity in Helpfulness Votes: Over 90% of user reviews have zero helpfulness votes. This gives us very little statistical information to work with for newly added reviews.

4. Surface Level Text Features analysis: Our model only extracts surface level metrics like word coumts, punctuation and does not consident Bangla language emotion analysis.

5. No User Account History: The model determines each review independently without trackiung user history of purchases and other reviews as long term customers or repeated customers may have some effects on the final review on e-commerce platforms.

9. Conclusion & Project Accomplishments

In conclusion, the research problem of this project has been solved since the analuysis proved the possibility of distinguishing verified from non verified Bangla book reviews on

Rokomari.com using linguistic and behavioral metrics. By solving severe class imbalance and target label skew issues, the findings from our analysis justified the project objective and showed that user-generated metadata can be effectively used to automatically assess the trustworthiness of reviews on Bangladeshi e-commerce platforms.

The main achievements obtained as a result of finishing this project show a number of important results. Firstly, hypothesis testing (Z-test Z = -15.5289, T-test T = -41.1466, ANOVA F = 115.9362) together with outlier analysis showed that removing the extreme outliers with respect to word count greatly decreased sample variance and also revealed significant differences in the behaviour of reviewers. Secondly, the use of Grid Search for hyperparameter tuning avoided both overfitting and severe underfitting, identifying the optimal parameters (Alpha* = 0.10, Lambda* = 0.00) which achieved a significant improvement on the baseline constant predictors (58.22% accuracy). Thirdly, calibration of the decision threshold (T* = 0.5899) was found to be essential for correcting class bias, since it increased Precision to 69.73%, raised Specificity to 64.26%, and reduced the Balanced Error Rate to 38.34%.

While complete accuracy is unrealistic with noisy human text because of natural behavioral overlap, these results show the practical value of our project for Rokomari.com. Adjusting the decision threshold ensures that when the system marks a purchase as verified, it does so accurately nearly 70% of the time. It also successfully identifies and filters out over 64% of unverified reviews. These achievements confirm that language and behavior signals can effectively predict outcomes for automated review verification in real-world recommendation systems..

10. References & Resources Cited

1. RokomariBG Dataset: E-Commerce Book Reviews and Metadata Corpus, Rokomari.com, Bangladesh.

11. Youtube Video Link

https://youtu.be/2Vwfsahp_zI?si=UjhF6AhJAoOKMdFD

12. Google Colab Notebook Link

https://drive.google.com/file/d/1ms7Pzr4DudF3fmB_C1y-t2SaR_IHAukx/view?usp=sharing